JAVA PROGRAMMING

6-Month Complete Learning Roadmap

From absolute beginner to Core Java, problem-solving, SQL, and a beginner backend project

Duration	6 months / 24 weeks / 168 study days
Suggested pace	1.5-2 hours per day, 6 days per week
Learning cycle	Concepts -> examples -> independent coding -> review -> projects
Expected outcome	Strong foundations and practical beginner-level skills; mastery requires continued practice.

How to use this roadmap

Treat each day as a target, not a race. If a concept is unclear, spend extra time practicing before moving forward. Maintain a code folder, a learning journal, and a list of questions. The plan is broad; advanced topics deepen through implementation and continued study.

Month 1 - Java Fundamentals

Days 1-28 | Build basic programming fluency: syntax, variables, I/O, operators, conditions, loops, methods, arrays, strings.

Week 1: Getting started

• D1: Java overview, features, JDK/JRE/JVM, install JDK and IDE, Hello World

• D2: Program structure, class, main method, statements, comments, compile/run

• D3: Variables, primitive types, constants, initialization

• D4: Output and Scanner input

• D5: Arithmetic, relational, logical, assignment, increment/decrement operators

• D6: Type conversion, casting, integer division

• D7: Revision and beginner assessment

Week 2: Decisions and loops

• D8: if, if-else, nested if

• D9: else-if ladder and switch

• D10: for loop, counters, sums, multiplication tables

• D11: while and do-while

• D12: Nested loops and pattern logic

• D13: break, continue, return basics

• D14: Weekly coding test

Week 3: Methods, arrays, strings

• D15: Methods, parameters, arguments, return values, void

• D16: Method calls, overloading, local scope

• D17: 1D arrays, indexing, length, traversal

• D18: Array sum, min/max, search

• D19: 2D arrays and matrix operations

• D20: String basics and common methods

• D21: Revision and assessment

Week 4: Practice and project

• D22: Number problems - prime, factorial, reverse, palindrome

• D23: Pattern practice and loop tracing

• D24: Array/string problem set

• D25: Plan Student Grade Calculator

• D26: Implement inputs, total, average, grade

• D27: Validate, support multiple students, debug

• D28: Month assessment

Practice / deliverable: Deliverable: console Student Grade Calculator. Aim for 40+ small programs. Exit goal: write and explain basic programs independently.

Month 2 - OOP and Intermediate Java

Days 29-56 | Design programs with classes and objects; learn OOP principles, recursion, and core language features.

Week 1: OOP foundations

- D29: Procedural vs OOP, core terminology
- D30: Classes, objects, fields, methods
- D31: Instance members and object state
- D32: Default and parameterized constructors
- D33: this keyword and constructor overloading
- D34: static fields, methods, initialization blocks
- D35: Review and practice

Week 2: OOP pillars

- D36: Encapsulation, private fields, getters/setters
- D37: Inheritance and inheritance structures
- D38: super and constructor chaining
- D39: Overriding and runtime polymorphism
- D40: Overloading vs overriding
- D41: Abstract classes and abstraction
- D42: Interfaces and OOP test

Week 3: Arrays, strings, recursion

- D43: Array traversal, copying, enhanced for
- D44: 2D arrays and matrices
- D45: Jagged arrays and arrays of objects
- D46: String immutability, pool, == vs equals
- D47: StringBuilder and StringBuffer
- D48: Recursion, base case, call stack
- D49: Recursion exercises

Week 4: Language features and project

- D50: Packages, imports, access modifiers
- D51: final, references, pass-by-value
- D52: Wrapper classes, boxing/unboxing
- D53: Enums, varargs, command-line arguments
- D54: Debugging and mixed challenges
- D55: Build Student Management System
- D56: Test and document

Practice / deliverable: Deliverable: console Student Management System with add/list/search/update/delete using in-memory data. Aim for 50+ exercises.

Month 3 - Collections and Modern Core Java

Days 57-84 | Handle errors, use data structures, write generic code, and apply lambdas and streams.

Week 1: Exception handling

- D57: Errors vs exceptions; hierarchy
- D58: try, catch, multiple catch
- D59: finally and execution flow
- D60: throw and throws
- D61: Checked and unchecked exceptions
- D62: Custom exceptions and propagation
- D63: Review and assessment

Week 2: Collections Framework

- D64: Collection vs Collections; overview
- D65: ArrayList operations
- D66: LinkedList, Vector, Stack overview
- D67: HashSet, LinkedHashSet, TreeSet
- D68: Queue, PriorityQueue, Deque
- D69: HashMap, LinkedHashMap, TreeMap
- D70: Iterator, ListIterator, traversal

Week 3: Generics and sorting

- D71: Generic classes and type parameters
- D72: Generic methods
- D73: Bounded types and wildcards
- D74: Comparable and natural ordering
- D75: Comparator and custom sorting
- D76: Lambda expressions
- D77: Functional interfaces and test

Week 4: Streams and project

- D78: Method references and functional interfaces
- D79: Stream filter, map, forEach
- D80: sorted, distinct, reduce, collect
- D81: Optional and null handling
- D82: Date/Time API: LocalDate, LocalTime, LocalDateTime
- D83: Build Employee Management System
- D84: Test and revise

Practice / deliverable: Deliverable: Employee Management System with collection-based storage, search, sorting, filters, and exception handling. Aim for 60+ exercises.

Month 4 - Advanced Core Java and JDBC

Days 85-112 | Learn file I/O, concurrency, JVM fundamentals, networking basics, and database connectivity.

Week 1: File I/O

- D85: Java I/O, File class, paths
- D86: Create, inspect, rename, delete files/directories
- D87: FileReader and FileWriter
- D88: BufferedReader and BufferedWriter
- D89: FileInputStream and FileOutputStream
- D90: Serialization/deserialization
- D91: File I/O assessment

Week 2: Multithreading

- D92: Processes, threads, concurrency
- D93: Thread class; start vs run
- D94: Runnable and thread lifecycle
- D95: sleep, join, interrupt
- D96: Race conditions, synchronization, thread safety
- D97: wait/notify/notifyAll
- D98: ExecutorService and thread pools

Week 3: JVM and networking

- D99: JVM architecture, class loading, bytecode
- D100: Runtime memory: stack, heap, metaspace overview
- D101: Garbage collection and memory leaks
- D102: Built-in/custom annotations
- D103: Reflection fundamentals
- D104: IP, ports, TCP/UDP, URL
- D105: Socket and ServerSocket basics

Week 4: JDBC and capstone

- D106: JDBC architecture, drivers, database setup
- D107: Connection, Statement, PreparedStatement
- D108: JDBC CRUD
- D109: Transactions, commit, rollback
- D110: Layered architecture and separation of concerns
- D111: Build Library Management System
- D112: Test, document, GitHub preparation

Practice / deliverable: Capstone: Library Management System with books, members, issue/return, overdue rules, JDBC, and a relational database. Aim for 40+ exercises.

Month 5 - DSA and SQL

Days 113-140 | Develop algorithmic thinking and database skills for practical Java development.

Week 1: Complexity and arrays

- D113: Big-O time and space complexity
- D114: Linear and binary search
- D115: Bubble, selection, insertion sort
- D116: Merge sort and quicksort
- D117: Two pointers and sliding window basics
- D118: Prefix sums and frequency counting
- D119: Timed problem set

Week 2: Linear structures

- D120: Linked list concepts and implementation
- D121: Singly/doubly linked list operations
- D122: Stack implementation and applications
- D123: Queue, circular queue, deque
- D124: Hashing patterns
- D125: Recursion and backtracking introduction
- D126: Assessment

Week 3: Trees and patterns

- D127: Binary tree terminology and traversals
- D128: Binary Search Tree operations
- D129: Heap and PriorityQueue
- D130: Graph concepts and representations
- D131: BFS and DFS
- D132: Greedy and dynamic programming introduction
- D133: Mixed problem-solving assessment

Week 4: SQL and integration

- D134: Relational databases, tables, keys
- D135: SELECT, WHERE, ORDER BY, LIMIT
- D136: INSERT, UPDATE, DELETE, constraints
- D137: JOINs, GROUP BY, aggregates
- D138: Subqueries and indexes overview
- D139: Java + SQL mini task or coding portfolio
- D140: Assessment and review

Practice / deliverable: Deliverable: problem-solving repository and SQL practice file. Target 70-100 algorithm problems at a sustainable pace; focus on understanding, not speed.

Month 6 - Java Application Development

Days 141-168 | Apply Java, databases, testing, build tools, and backend fundamentals in a structured application.

Week 1: Developer workflow

- D141: Git/GitHub status, add, commit, push, branches
- D142: Maven structure, dependencies, lifecycle
- D143: JUnit and unit testing
- D144: Debugging, logging concepts, clean code
- D145: HTTP, JSON, REST concepts
- D146: Spring and dependency injection overview
- D147: Spring Boot setup and review

Week 2: Spring Boot fundamentals

- D148: Controllers, services, repositories, layers
- D149: REST endpoints and request mapping
- D150: DTOs, request bodies, path/query parameters
- D151: Validation and API exception handling
- D152: Connect Spring Boot to relational database
- D153: JPA entities and repositories
- D154: CRUD endpoints and tests

Week 3: Capstone build

- D155: Requirements, entities, API plan
- D156: Data model and schema
- D157: Create/read endpoints
- D158: Update/delete and validation
- D159: Search/filter and business rules
- D160: Unit and API tests
- D161: Integration review and bug fixes

Week 4: Finish and present

- D162: Error responses and logging
- D163: README, setup guide, API examples
- D164: Security basics: secrets, validation, password hashing concepts
- D165: Refactor and code quality
- D166: Final testing and sample data
- D167: Publish repository and prepare demo
- D168: Final assessment and next plan

Practice / deliverable: Capstone: Spring Boot REST API with relational database, CRUD, validation, exception handling, tests, and documentation. This is a beginner portfolio project; continue learning after the six-month plan.

Final 6-Month Completion Checklist

■ Month 1: Write programs using variables, I/O, conditions, loops, methods, arrays, and strings.

■ Month 2: Use classes, objects, constructors, inheritance, interfaces, and polymorphism.

■ Month 3: Apply exceptions, collections, generics, sorting, lambdas, streams, and date/time APIs.

■ Month 4: Work with files, threads, JVM concepts, networking basics, and JDBC.

■ Month 5: Explain Big-O and implement common structures, algorithms, and SQL queries.

■ Month 6: Build, test, document, and publish a beginner Spring Boot REST API.

Suggested weekly rhythm

Days	Activity
1-4	Learn concepts, trace examples, and write small programs
5	Independent exercises and debugging
6	Weekly review, quiz, and project progress
7	Rest or optional light revision

Completing this plan is a milestone, not a claim of mastery. Keep solving problems, revisiting fundamentals, and extending your projects.